MSG No. 195.2 $P231'$ [Type II, cubic]

Table 1: Wyckoff site: 1a, site symmetry: 23.1'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 1b, site symmetry: 23.1'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: 3c, site symmetry: 222..1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,8,10,12,17,20,22,24]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,7,9,11,18,19,21,23]

Table 4: Wyckoff site: 3d, site symmetry: 222..1'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,2,3,4,13,14,15,16]
2	$\left[0, \frac{1}{2}, 0\right]$	[5,8,10,12,17,20,22,24]
3	$\left[0, 0, \frac{1}{2}\right]$	[6,7,9,11,18,19,21,23]

Table 5: Wyckoff site: 4e, site symmetry: .3.1'

No.	position	mapping
1	$[x, x, x]$	[1,5,6,13,17,18]
2	$[x, -x, -x]$	[2,10,11,14,22,23]
3	$[-x, x, -x]$	[3,7,12,15,19,24]
4	$[-x, -x, x]$	[4,8,9,16,20,21]

Table 6: Wyckoff site: 6f, site symmetry: 2..1'

No.	position	mapping
1	$[x, 0, 0]$	[1,2,13,14]
2	$[-x, 0, 0]$	[3,4,15,16]
3	$[0, x, 0]$	[5,12,17,24]
4	$[0, 0, x]$	[6,9,18,21]
5	$[0, 0, -x]$	[7,11,19,23]
6	$[0, -x, 0]$	[8,10,20,22]

Table 7: Wyckoff site: 6g, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{2}\right]$	[1,2,13,14]
2	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[3,4,15,16]
3	$\left[\frac{1}{2},\, x,\, 0\right]$	[5,12,17,24]
4	$\left[0,\, \frac{1}{2},\, x\right]$	[6,9,18,21]
5	$\left[0,\, \frac{1}{2},\, -x\right]$	[7,11,19,23]
6	$\left[\frac{1}{2},\, -x,\, 0\right]$	[8,10,20,22]

Table 8: Wyckoff site: 6h, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,2,13,14]
2	$\left[-x,\, \frac{1}{2},\, 0\right]$	[3,4,15,16]
3	$\left[0,\, x,\, \frac{1}{2}\right]$	[5,12,17,24]
4	$\left[\frac{1}{2},\, 0,\, x\right]$	[6,9,18,21]
5	$\left[\frac{1}{2},\, 0,\, -x\right]$	[7,11,19,23]
6	$\left[0,\, -x,\, \frac{1}{2}\right]$	[8,10,20,22]

Table 9: Wyckoff site: 6i, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[1,2,13,14]
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[3,4,15,16]
3	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[5,12,17,24]
4	$\left[\frac{1}{2},\, \frac{1}{2},\, x\right]$	[6,9,18,21]
5	$\left[\frac{1}{2},\, \frac{1}{2},\, -x\right]$	[7,11,19,23]
6	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[8,10,20,22]

Table 10: Wyckoff site: 12j, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,13]
2	$[x,\, -y,\, -z]$	[2,14]
3	$[-x,\, y,\, -z]$	[3,15]
4	$[-x,\, -y,\, z]$	[4,16]
5	$[z,\, x,\, y]$	[5,17]
6	$[y,\, z,\, x]$	[6,18]
7	$[-y,\, z,\, -x]$	[7,19]
8	$[-z,\, -x,\, y]$	[8,20]
9	$[-y,\, -z,\, x]$	[9,21]
10	$[z,\, -x,\, -y]$	[10,22]
11	$[y,\, -z,\, -x]$	[11,23]
12	$[-z,\, x,\, -y]$	[12,24]